Directed Linear Reinforcement Is an Independent-Row Dirichlet Environment

Source-local theorem package HCS-C342

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Abstract

For every finite strongly connected directed multigraph with a nonempty outgoing row at each vertex and positive weights on labelled arcs, we give a convention-complete theorem for linear directed edge reinforcement. Its exact path law is a product of vertex-wise rising factorials and equals the annealed law of independent Dirichlet transition rows. We derive the posterior after every history and prove almost-sure transition, vertex-occupation and labelled-edge-occupation limits. Loops, parallel arcs, deterministic rows and the one-vertex Pólya boundary are included. Undirected ERRW, infinite graphs, arithmetic targets and Route B are excluded.

1 Frozen model

Let $G = (V, E)$ be a finite strongly connected directed multigraph in which every vertex has at least one outgoing arc. Each arc is labelled, including parallel arcs, and has $\alpha_e > 0$. Write $\alpha_v = \sum_{e:e^- = v} \alpha_e > 0$. Starting at x_0, define labelled arc and departure counts by $N_e(t)$ and $N_v(t) = \sum_{e:e^-=v} N_e(t)$. From $v = X_t$ choose e with $e^- = v$ according to

$$\mathbb{P}(e \mid \mathcal{F}_t) = \frac{\alpha_e + N_e(t)}{\alpha_v + N_v(t)}. \tag{1}$$

Only the augmented process containing the counts is asserted to be Markov.

Theorem 1 (Mixture, posterior and learning)**.** *For every legal labelled arc path $\gamma = (e_1, \ldots, e_m)$,*

$$\mathbb{P}(\gamma) = \prod_{v \in V} \frac{\prod_{e:e^-=v} (\alpha_e)_{N_e(\gamma)}}{(\alpha_v)_{N_v(\gamma)}}. \tag{2}$$

Independently sample each row

$$\omega(v, \cdot) \sim \mathrm{Dirichlet}((\alpha_e)_{e^-=v}). \tag{3}$$

Then the reinforced path law equals the annealed law of the Markov chain that, conditional on ω, selects labelled arc e with probability ω_e. After any history γ, the rows remain independent and

$$\omega(v, \cdot) \mid \gamma \sim \mathrm{Dirichlet}((\alpha_e + N_e(\gamma))_{e^-=v}). \tag{4}$$

Almost surely every vertex is visited infinitely often and

$$\frac{N_e(t)}{N_{e^-}(t)} \to \omega_e, \qquad \frac{N_v(t)}{t} \to \pi_\omega(v), \qquad \frac{N_e(t)}{t} \to \pi_\omega(e^-)\omega_e, \tag{5}$$

where π_ω is the unique stationary law of $K_\omega(v, w) = \sum_{e:v \to w} \omega_e$.

2 Exact path law and mixture

At the r-th departure from v, the denominator in (1) is $\alpha_v + r$. When arc e is selected for the s-th time, its numerator is $\alpha_e + s$. Multiplication and grouping by departure vertex gives (2), including loops and labelled parallel arcs. Thus legal paths with the same start and the same outgoing labelled counts at every vertex have equal probability.

For positive $(\alpha_1, \ldots, \alpha_d)$, direct integration of the Dirichlet density gives

$$\mathbb{E}\prod_i \omega_i^{n_i} = \frac{\Gamma(\sum_i \alpha_i)}{\Gamma(\sum_i \alpha_i + \sum_i n_i)} \prod_i \frac{\Gamma(\alpha_i + n_i)}{\Gamma(\alpha_i)} = \frac{\prod_i (\alpha_i)_{n_i}}{(\sum_i \alpha_i)_{\sum_i n_i}}. \tag{6}$$

Given ω, a path has probability $\prod_e \omega_e^{N_e(\gamma)}$. Independence of rows and (6) reproduce (2), proving equality of every finite-dimensional law without invoking an abstract representation theorem.

3 Source lineage

The directed reinforcement/RWRE correspondence is due to Enriquez and Sabot, *C. R. Math.* 335 (2002), 941–946, doi:10.1016/S1631-073X(02)02580-3. Partial exchangeability follows the lineage of Diaconis and Freedman, *Ann. Probab.* 8 (1980), 115–130, Project Euclid. Sabot and Tournier provide an authoritative overview, doi:10.5802/afst.1542. Our positive finite-graph specialization is reconstructed directly and makes no priority claim.